

Predictive Deconvolution in Practice



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Summary of Convolution: from core to seismogram

deconvolution ← → convolution

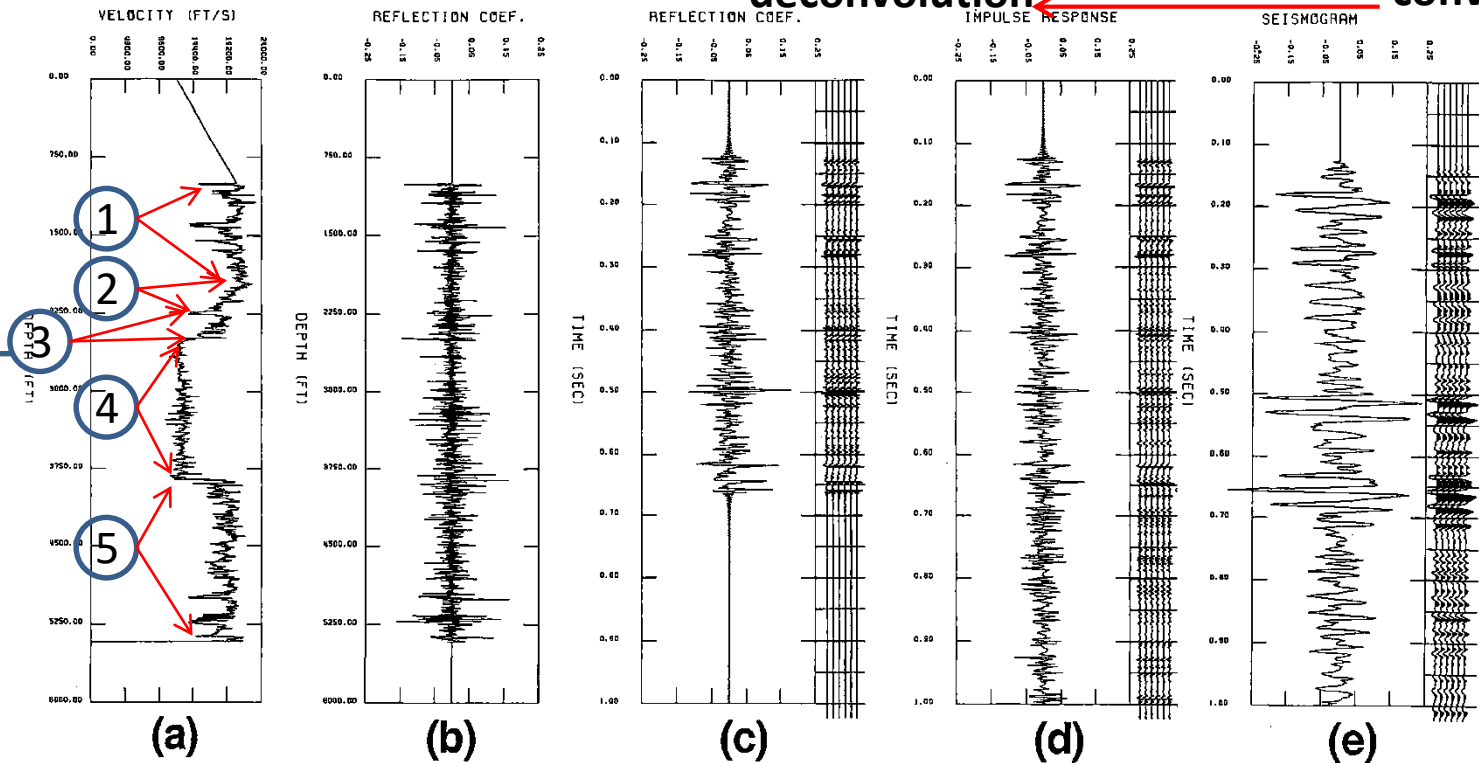


FIG. 2-8. (a) A segment of a measured sonic log, (b) the reflection coefficient series derived from (a), (c) the series in (b) after converting the depth axis to two-way time axis, (d) the impulse response that includes the primaries (c) and multiples, (e) the synthetic seismogram derived from (d) convolved with the source wavelet in Figure 2-11. One-dimensional seismic modeling means getting e from a . Deconvolution yields d and c , while 1-D inversion means getting a from d . Identify the event on a and b that corresponds to the big spike at 0.5 s in c . Impulse response d is a composite of the primaries c and all types of multiples.

Table 2-1. The interval velocity trend obtained from the sonic log in Figure 2-8a.

Layer Number	Interval Velocity, ft/s	Depth Range, ft
1	21000	1000–2000
2	19000*	2000–2250
3	18750	2250–2500
4	12650	2500–3775
5	19650	3775–5400

* The velocity in this layer gradually decreases.

Table 2-2. Stratigraphic identification associated with layering described in Table 2-1.

Layer Number	Lithologic Unit
1	Limestone
2	Shaly limestone with gradual increase in shale content
3	Shaly limestone
4	Sandstone
5	Dolomite

Review of Assumptions underlying Predictive Deconvolution

- Assumption 1a. The earth is made up of horizontal layers of constant velocity.
- Assumption 1b. The source generates a compressional plane wave that impinges on layer boundaries at normal incidence. Under such circumstances, no shear waves are generated.
- Assumption 2. The source waveform does not change as it travels in the subsurface; i.e., it is stationary.
- Assumption 3. The noise component $n(t)$ is zero.
- Assumption 4. Reflectivity is a random process. This implies that the seismogram has seismic wavelet characteristics since their autocorrelations and amplitude spectra are similar.
- Assumption 5. The seismic wavelet is minimum phase. Therefore, it has a minimum-phase inverse.

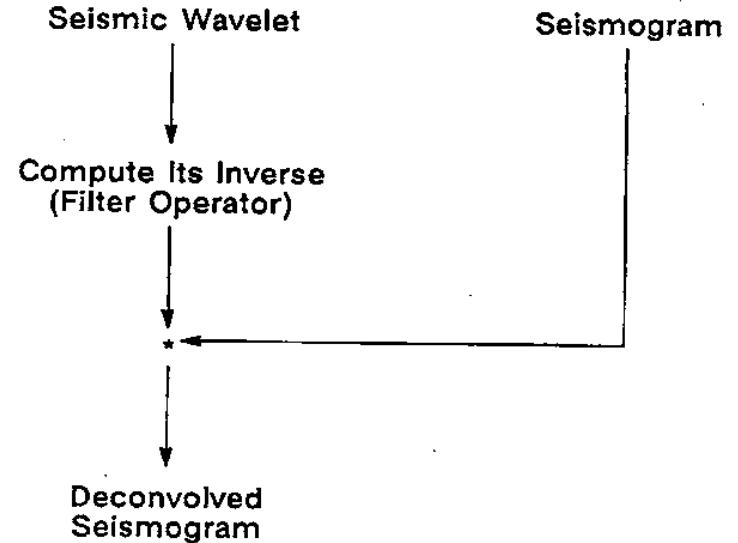


FIG. 2-14. A flowchart for inverse filtering.

Examples using known reflectivity (R)
compared to predicted reflectivity
from deconvolution (inverse filtering)
of seismogram ($S=W*R$)

Examples are shown for series R with
only a few reflectors (both widely and
closely spaced) and a more realistic
series with many random reflectors.

Perfect result when source is known, minimum phase W (Figs. 2-31c and 2-32c)
 Slightly degraded when W is unknown, minimum phase (Figs. 2-31d and 2-32d)

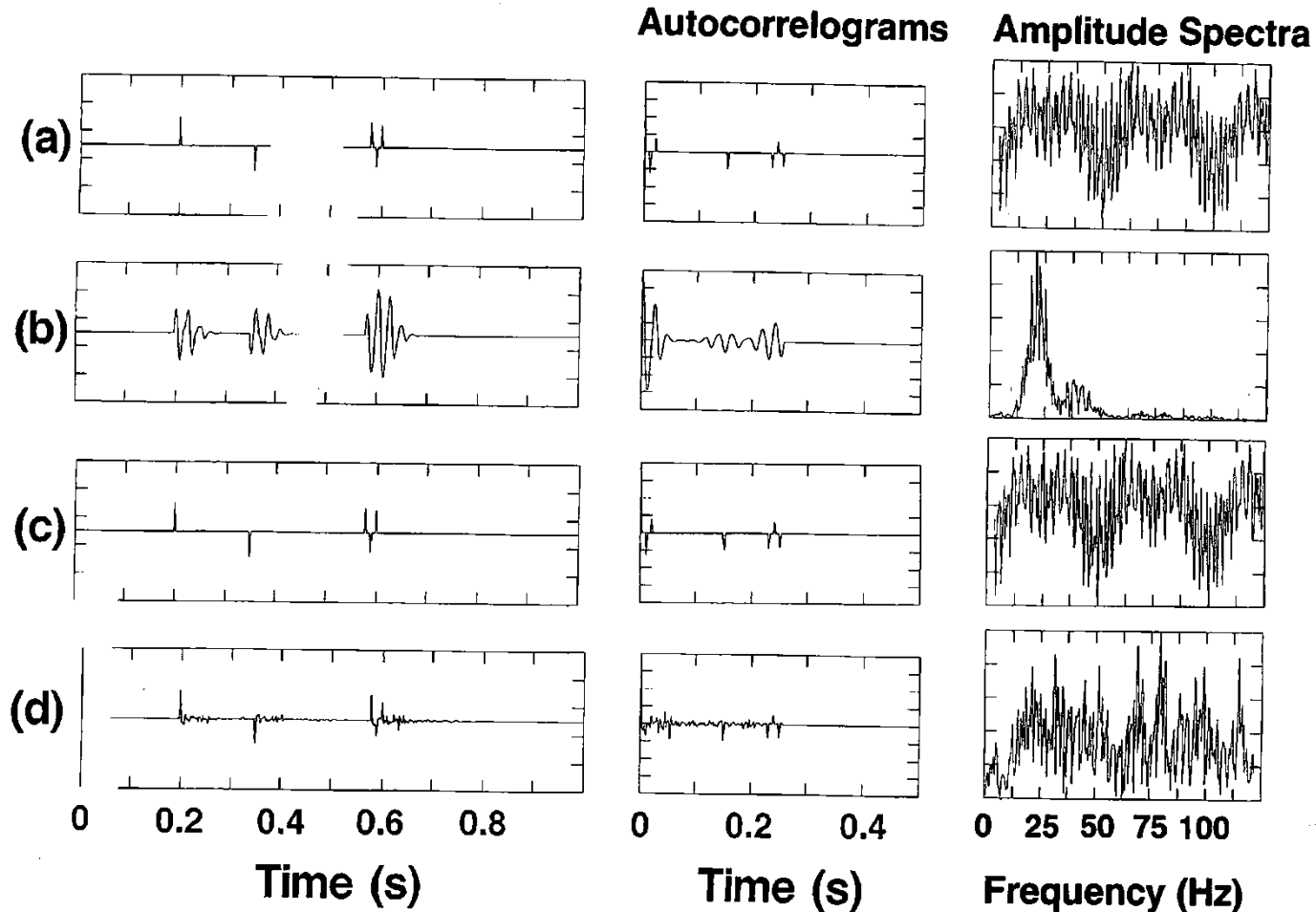


FIG. 2-31. (a) Impulse response, (b) seismogram, (c) spiking deconvolution using known, minimum-phase wavelet, (d) deconvolution assuming an unknown, minimum-phase source wavelet. Impulse response (a) is a sparse-spike series. For an unknown source wavelet (in violation of assumption 4), spiking deconvolution yields a less than perfect result. [Compare (c) and (d).]



Perfect result when source is known, minimum phase W (Figs. 2-31c and 2-32c)
 Slightly degraded when W is unknown, minimum phase (Figs. 2-31d and 2-32d)

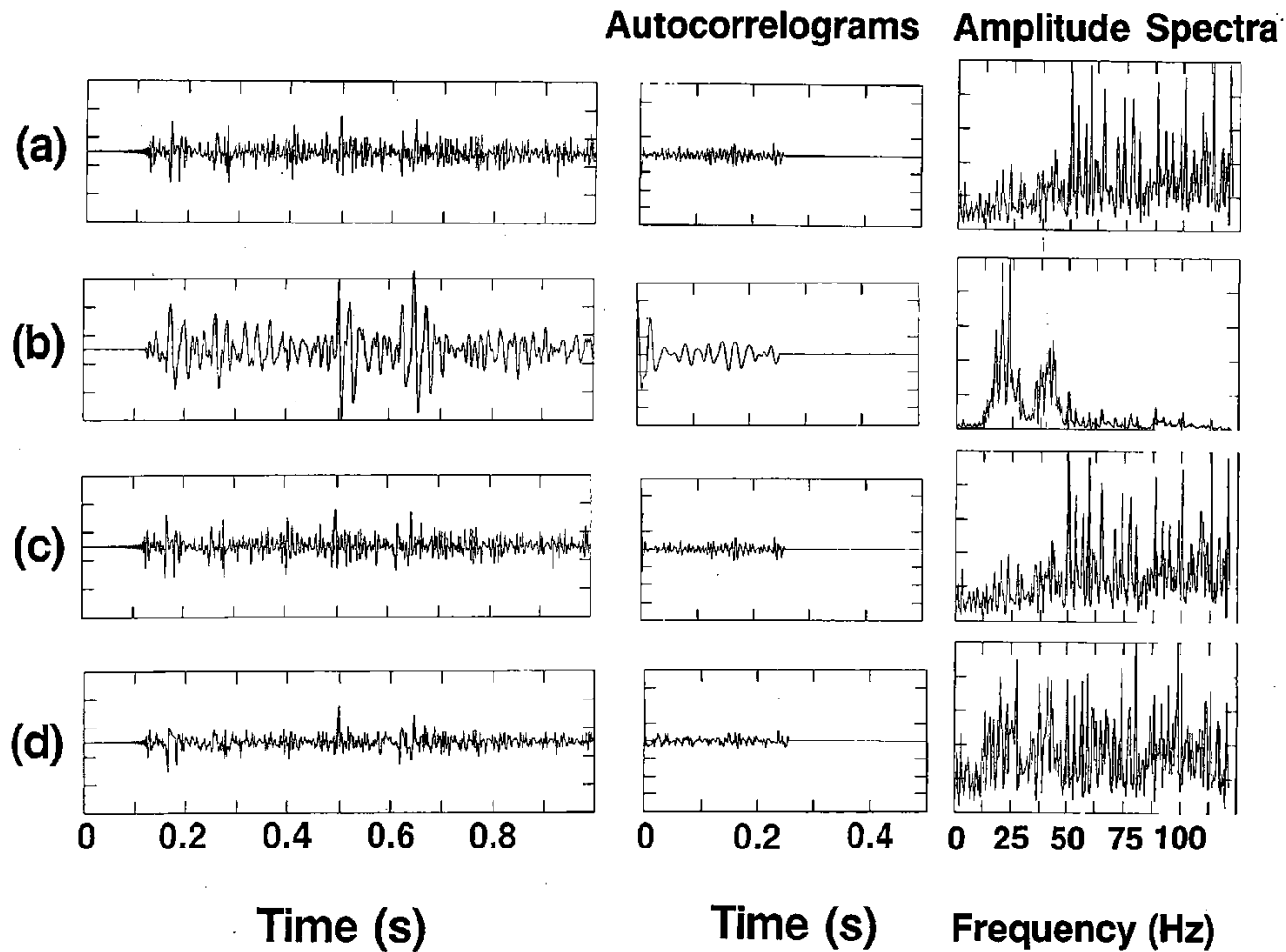


FIG. 2-32. (a) Impulse response, (b) seismogram, (c) spiking deconvolution using known, minimum-phase source wavelet, (d) deconvolution assuming an unknown, minimum-phase source wavelet. Impulse response (a) is based on a sonic log (Figure 2-8a). For the unknown source wavelet (in violation of assumption 4), spiking deconvolution yields a less than perfect result. [Compare (c) and (d).]



Much worse result for known, mixed phase W (Figs. 2-33c and 2-34c)
 Almost useless for unknown, mixed phase W (Figs. 2-33d and 2-34d)

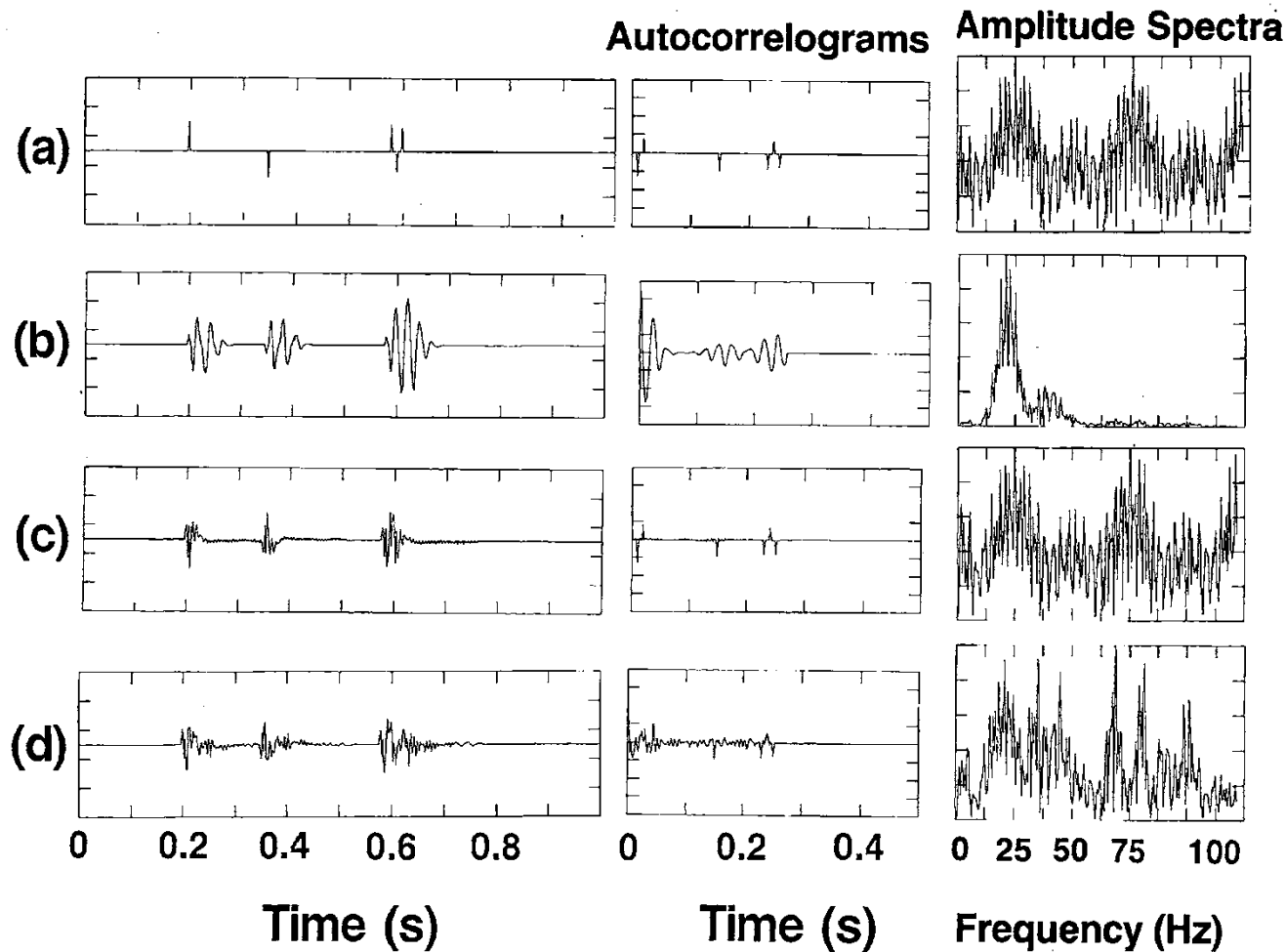


FIG. 2-33. (a) Impulse response, (b) seismogram, (c) deconvolution using a known, mixed-phase source wavelet, (d) deconvolution assuming an unknown, mixed-phase source wavelet. Impulse response (a) is a sparse-spike series. For a mixed-phase source wavelet (in violation of assumption 5), spiking deconvolution yields a degraded output (d), even when the wavelet is known (c).



Much worse result for known, mixed phase W (Figs. 2-33c and 2-34c)
 Almost useless for unknown, mixed phase W (Figs. 2-33d and 2-34d)

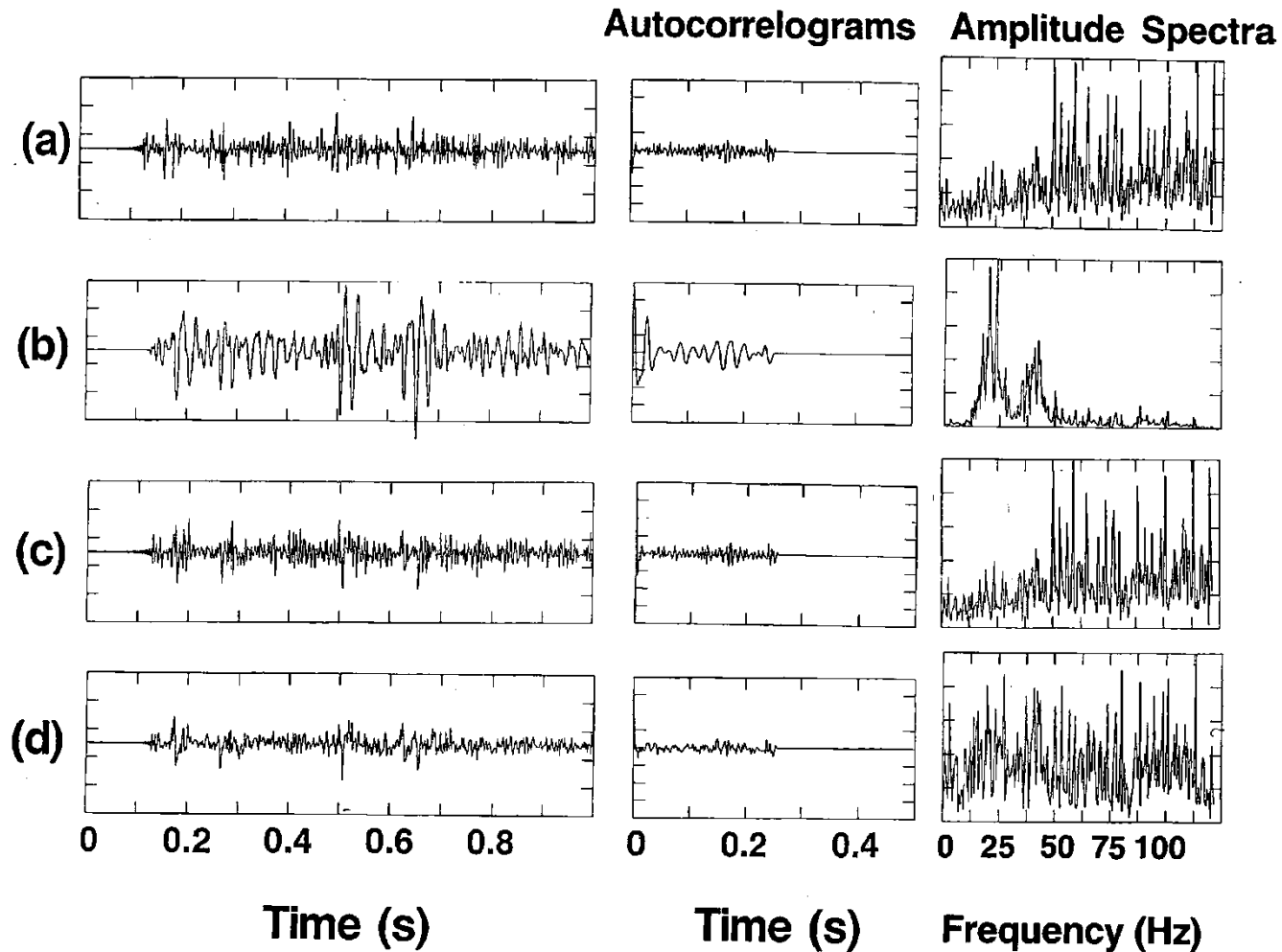


FIG. 2-34. (a) Impulse response, (b) seismogram, (c) deconvolution using a known, mixed-phase source wavelet, (d) deconvolution assuming an unknown, mixed-phase source. Impulse response (a) is based on the sonic log of Figure 2-8a. For the mixed-phase source wavelet (in violation of assumption 5), spiking deconvolution yields a degraded output (d) even when the wavelet is known (c).



Totally useless when noise is added to unknown, mixed phase W (Fig. 2-35c)

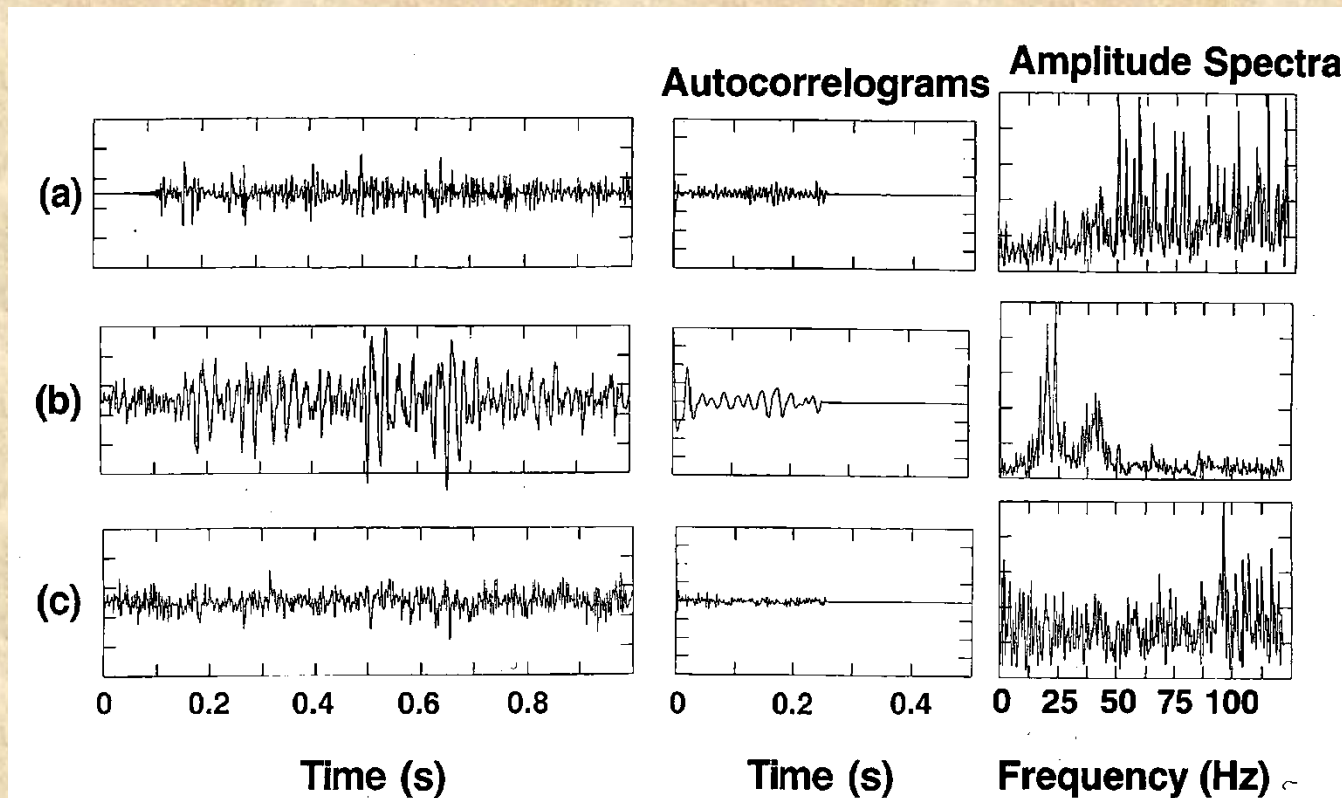


FIG. 2-35. (a) Impulse response, (b) seismogram with noise, (c) deconvolution assuming an unknown, mixed-phase source wavelet. Impulse response (a) is based on the sonic log of Figure 2-8a. In the presence of random noise (in violation of assumption 3), spiking deconvolution can produce a result with no relation to the earth's reflectivity [compare (a) to (c).]

Deconvolution as special examples of Wiener Optimum Filters (Fig. 2-30):

- Spiking deconvolution (Inverse Filtering) for zero or unit lag
- Predictive deconvolution (multiple removal)
- wavelet shaping (produce minimum phase W)

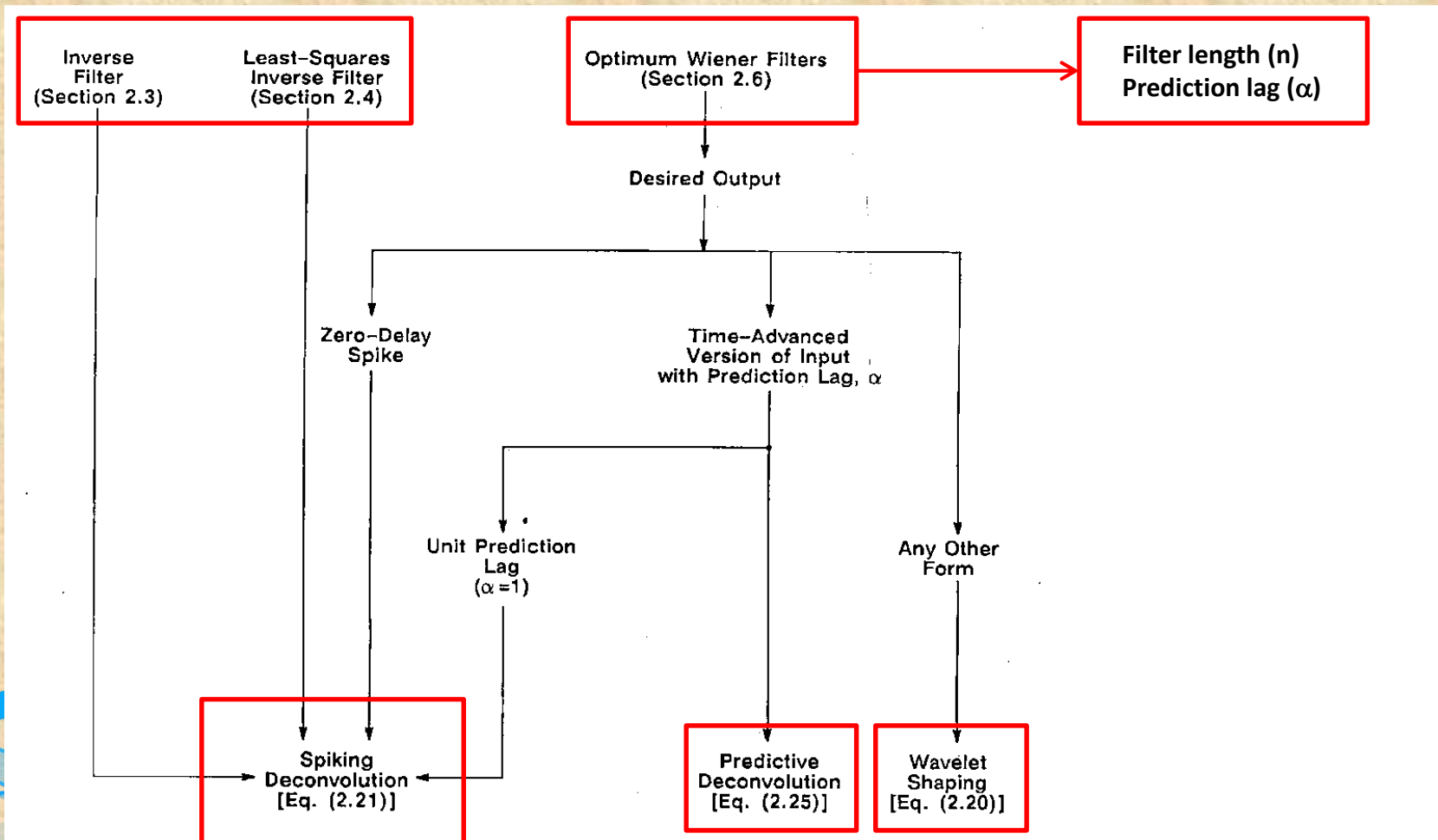


FIG. 2-30. A flowchart for interrelations between various deconvolution filters.



Considerations of filter length (n) and lag (gap) (α)



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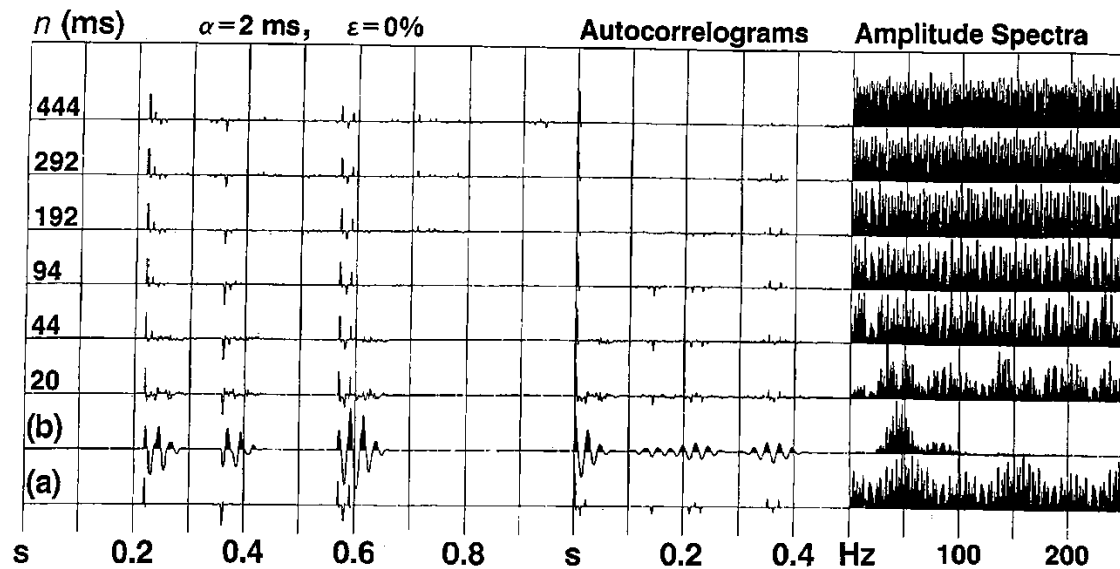


FIG. 2-38. Test of operator length where n = operator length, α = prediction lag, and ϵ = percent prewhitening. (a) Impulse response, (b) seismogram with unknown, minimum-phase source wavelet.

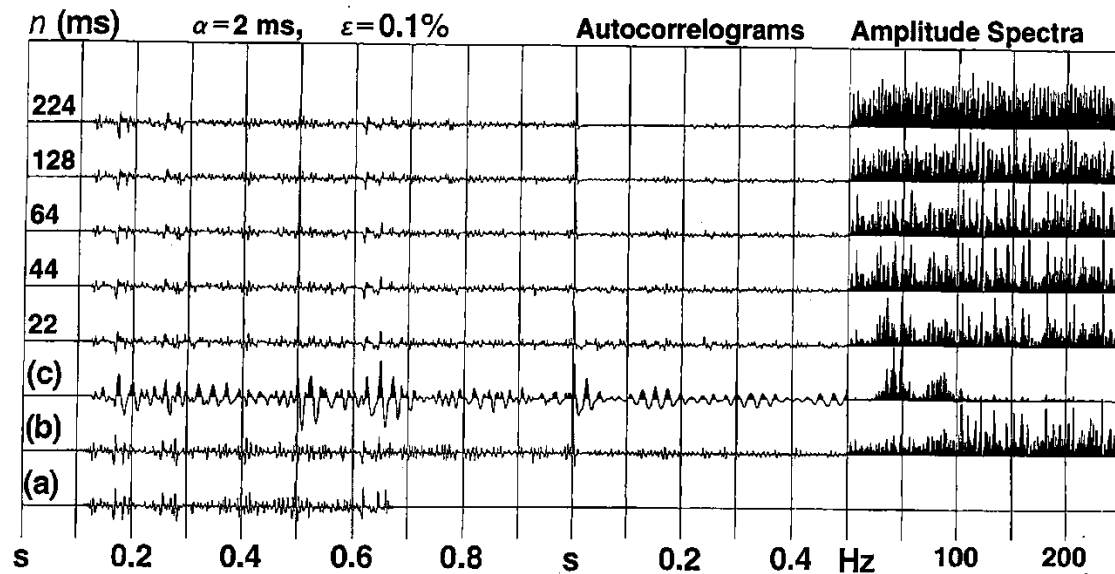


FIG. 2-39. Test of operator length where n = operator length, α = prediction lag, and ϵ = percent prewhitening. (a) Reflectivity, (b) impulse response, (c) seismogram with unknown, minimum-phase source wavelet.

Consideration of filter length (n) for spiking deconv (lag=2 ms)

Good result for unknown, minimum phase W when n is as long as W (e.g. 94 ms) (Figs. 2-38 and 2-39)

Longer filter lengths don't improve result very much



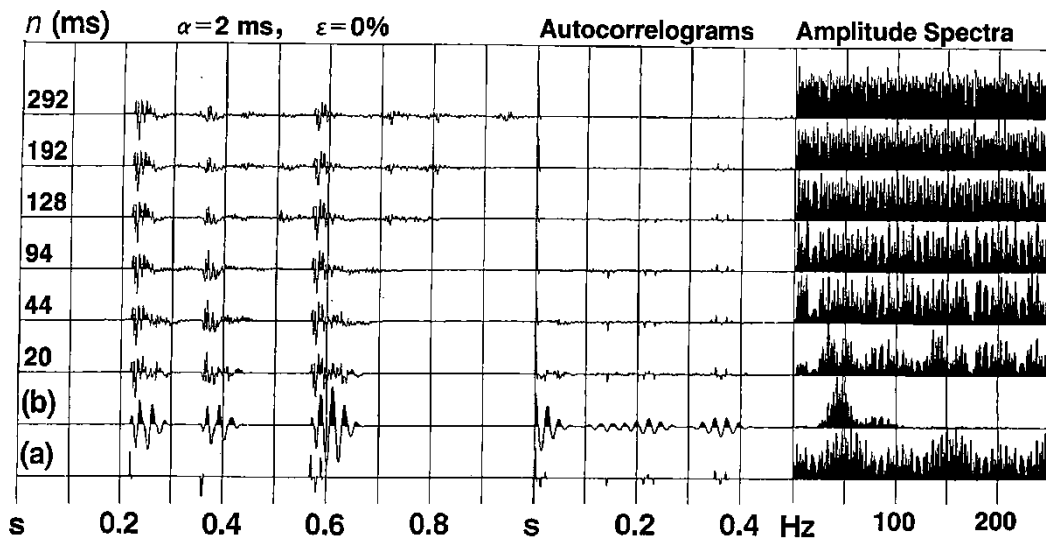


FIG. 2-42. Test of operator length where n = operator length, α = prediction lag, and ε = percent prewhitening. (a) Impulse response, (b) seismogram with unknown, mixed-phase source wavelet.

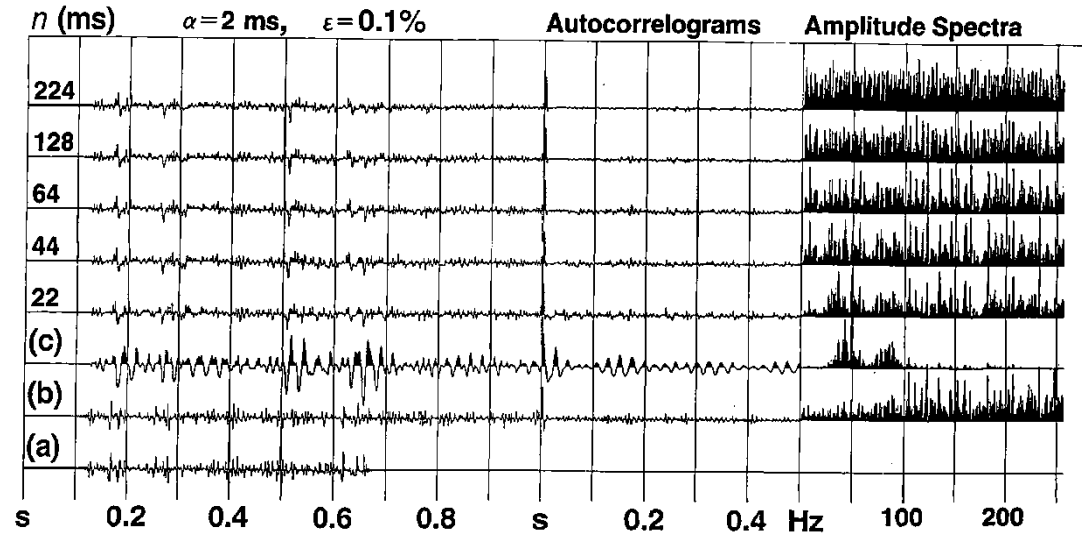


FIG. 2-43. Test of operator length where n = operator length, α = prediction lag, and ε = percent prewhitening. (a) Reflectivity, (b) impulse response, (c) seismogram with unknown, mixed-phase source wavelet.

Consideration of filter length (n)
for spiking deconv (lag=2 ms)

Worse for mixed phase W
(Figs. 2-42 and 2-43)



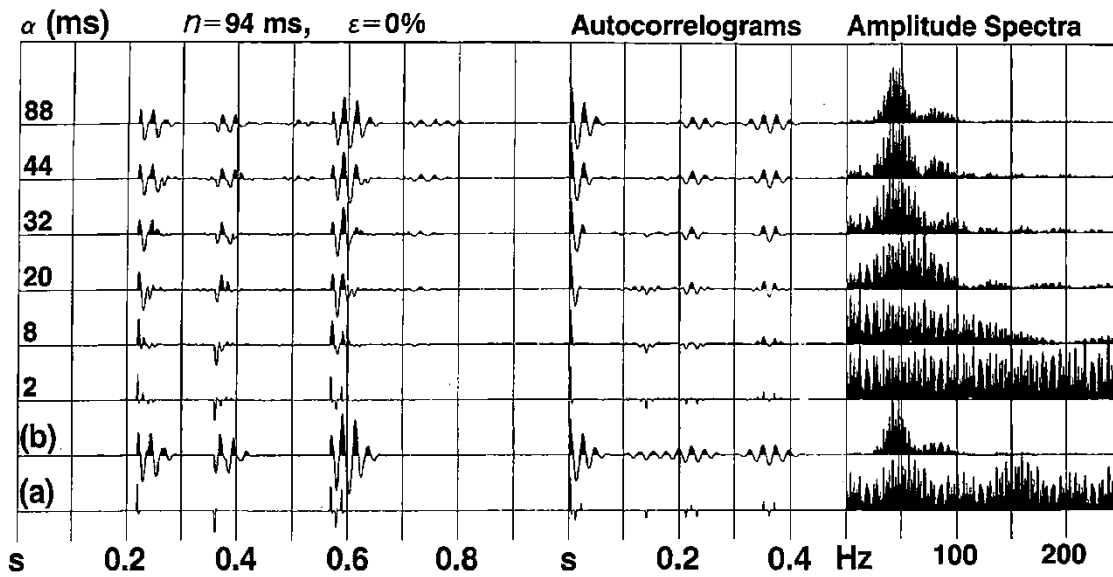


FIG. 2-46. Test of prediction lag where n = operator length, α = prediction lag, and ϵ = percent prewhitening. (a) Impulse response, (b) seismogram with unknown, minimum-phase source wavelet.

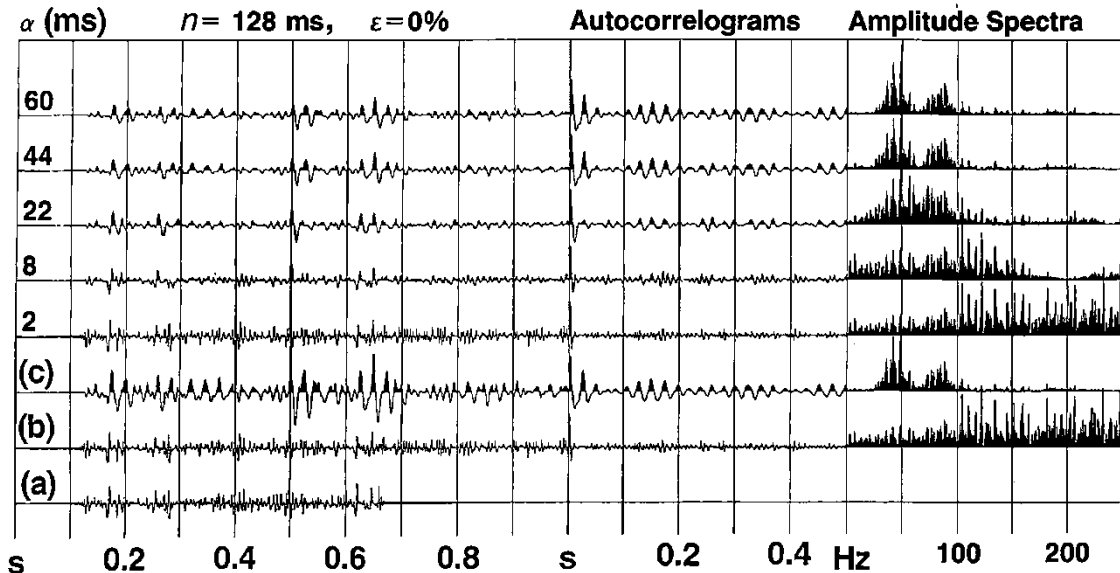


FIG. 2-47. Test of prediction lag where n = operator length, α = prediction lag, and ϵ = percent prewhitening. (a) Reflectivity, (b) impulse response, (c) seismogram with known minimum-phase source wavelet.

Tests of predictive lag

Spiking greatest for smallest lag

Large lag gives same results as original (Fig. 2-46)

Near perfect result for known, minimum phase W (Fig. 2-47)



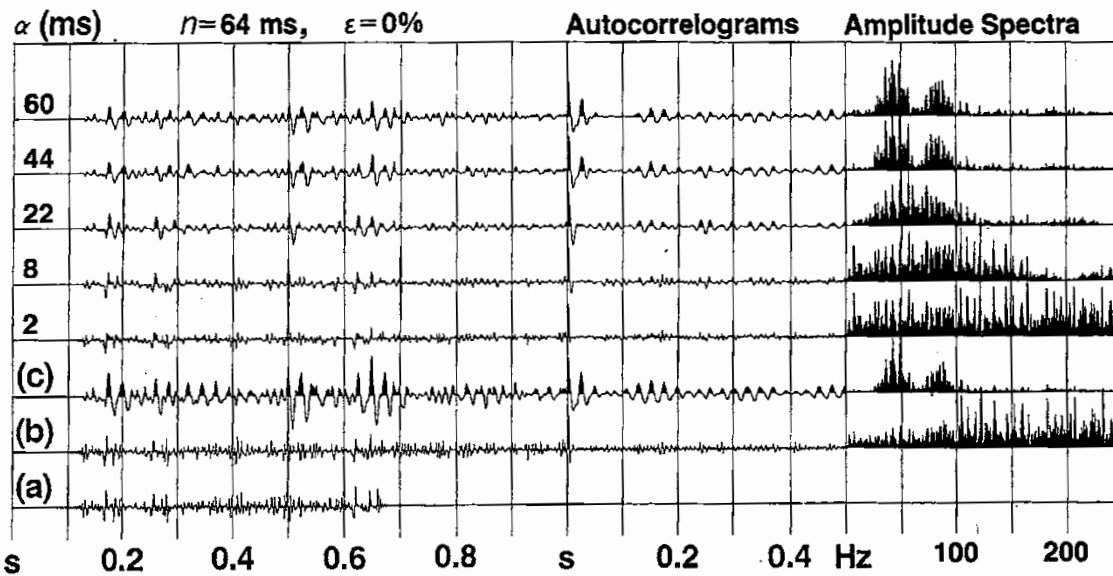


FIG. 2-48. Test of prediction lag where n = operator length, α = prediction lag, and ϵ = percent prewhitening. (a) Reflectivity, (b) impulse response, (c) seismogram with unknown, minimum-phase source wavelet.

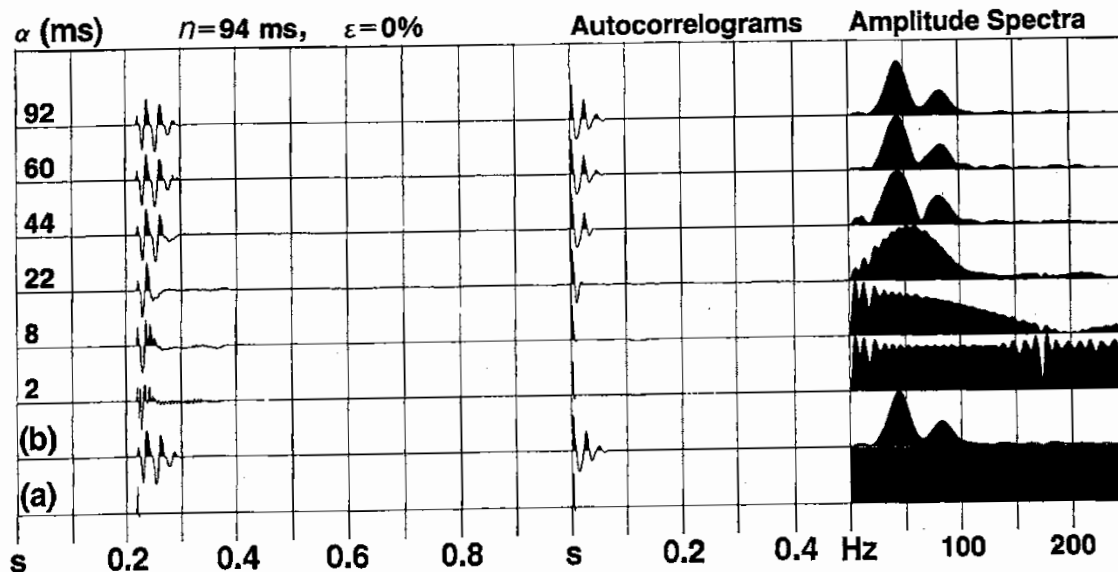


FIG. 2-49. Test of prediction lag for a single, isolated input wavelet where n = operator length, α = prediction lag, and ϵ = percent prewhitening. (a) Impulse response, (b) seismogram with mixed-phase source wavelet.

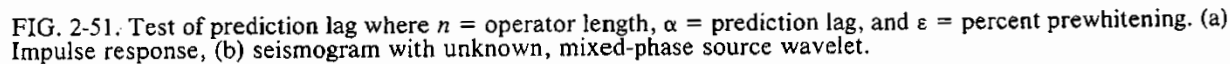
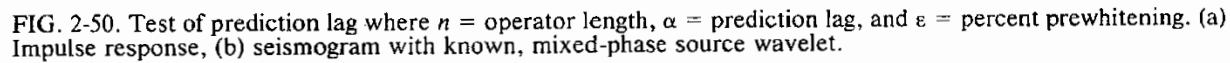
Tests of predictive lag

Adequate result for unknown, minimum phase W (Fig. 2-48)

Not very good for known, mixed phase W (Fig. 2-49)



Not very good for known,
mixed phase W (Figs 2-50)



Poor result for unknown, mixed phase W (Fig. 2-51)



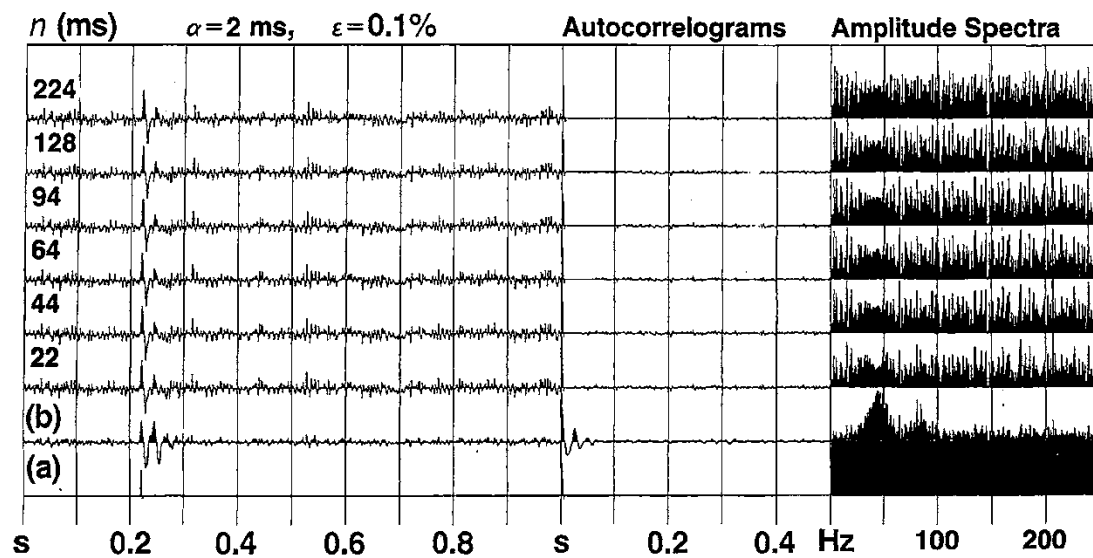


FIG. 2-61. Effect of random noise on deconvolution performance. Input seismogram (b) associated with reflectivity (a) contains a single, isolated wavelet (at around 0.2 s) buried in random noise. Here n = operator length, α = prediction lag, and ϵ = percent prewhitening.

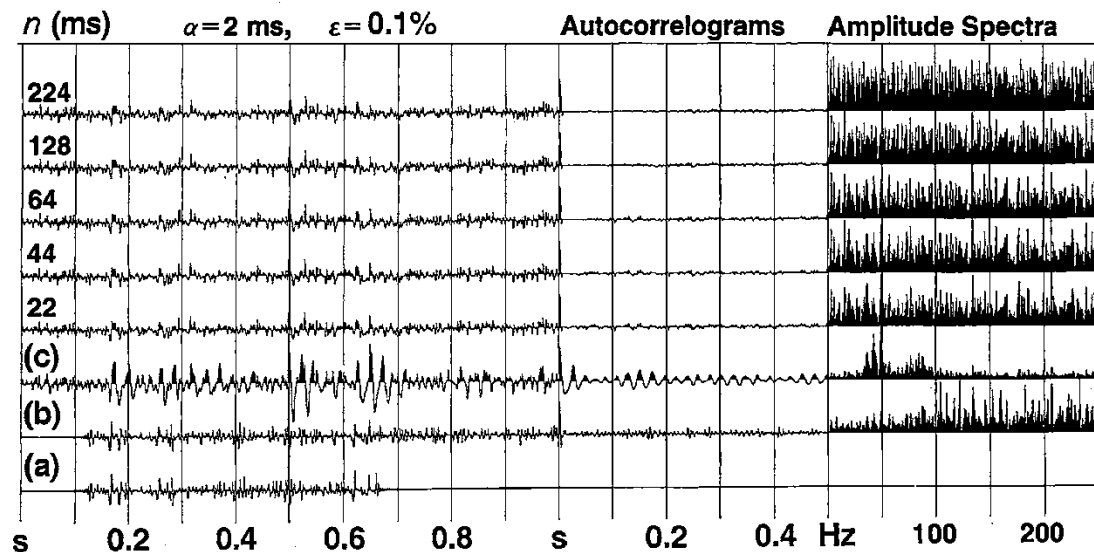


FIG. 2-62. Effect of random noise on deconvolution performance. (a) Reflectivity, (b) impulse response. Input seismogram (c) with unknown, minimum-phase source wavelet is noise contaminated. Here n = operator length, α = prediction lag, and ϵ = percent prewhitening.

When noise is added

Adequate if simple strong reflector (Fig. 2-61)

Worse if complex R and unknown, minimum phase W (Fig. 2-62)



When noise is added

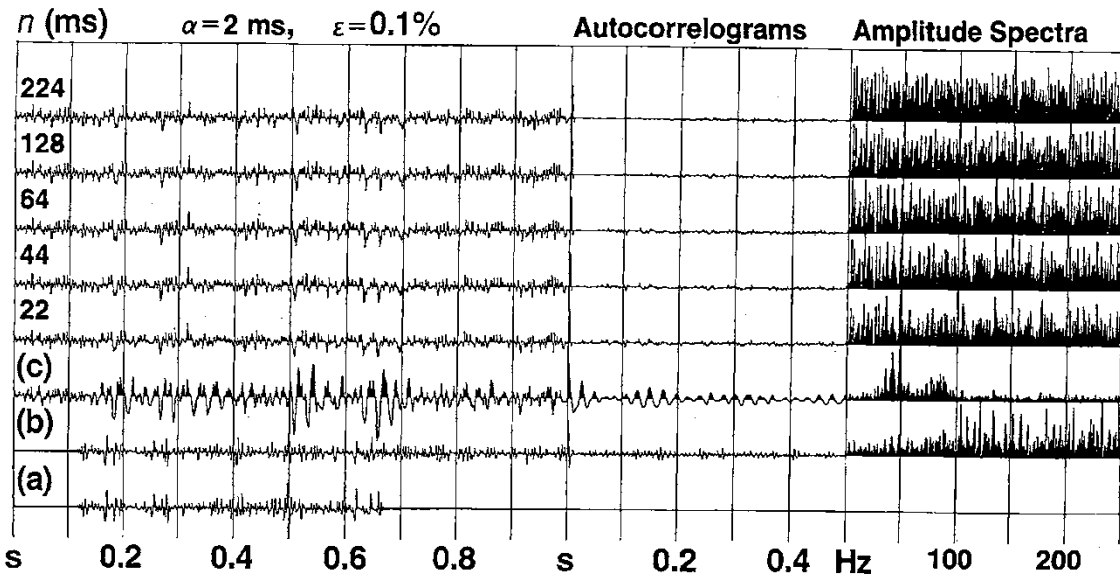


FIG. 2-63. Effect of random noise on deconvolution performance. (a) Reflectivity, (b) impulse response. Input seismogram (c) with unknown, mixed-phase source wavelet is noise contaminated. Here n = operator length, α = prediction lag, and ε = percent prewhitening.

Useless if unknown, mixed phase W and complex R (Fig. 2-63)

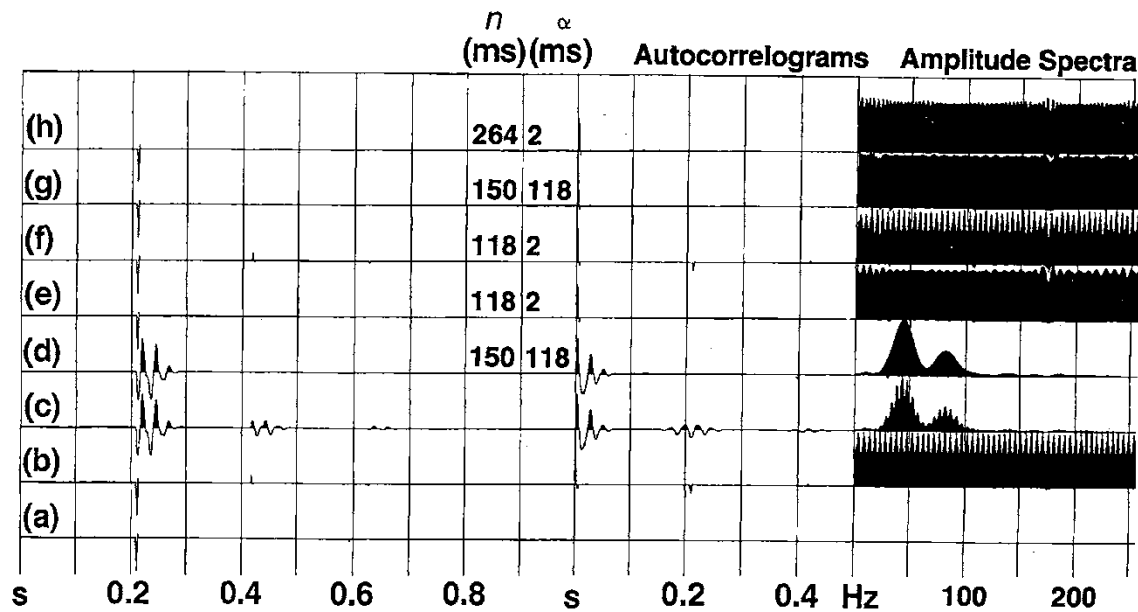


FIG. 2-64. (a) Reflectivity, (b) impulse response, (c) seismogram. Two-step deconvolution aimed at suppressing the multiple wave train, then spiking the remaining primary wavelet (d) to (e). The process can be performed in the reverse order (f) to (g). Here, n = operator length and α = prediction lag.

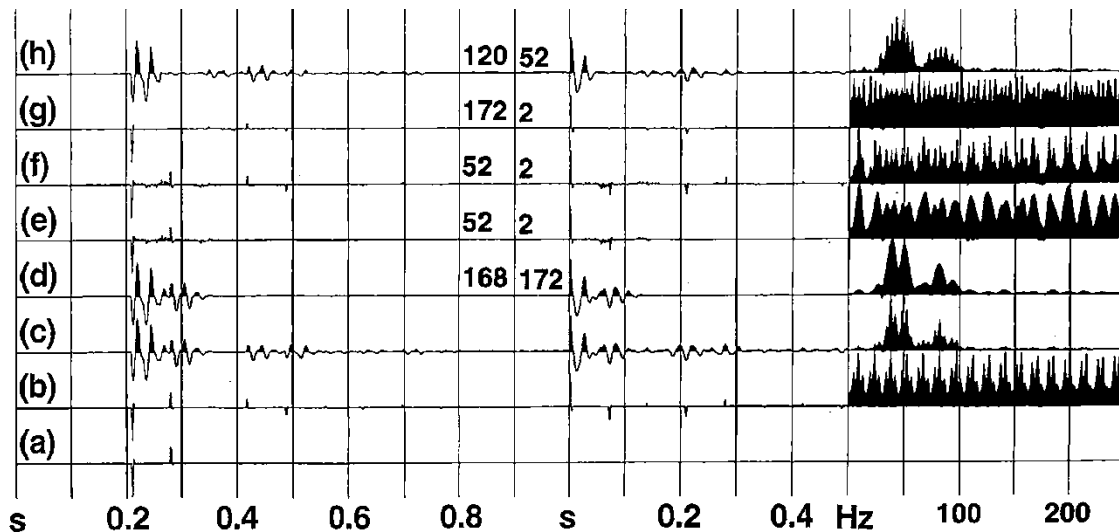


FIG. 2-65. Predictive deconvolution for multiple suppression. (a) Reflectivity, (b) impulse response, (c) seismogram. Two-step deconvolution: Predictive deconvolution (d) followed by spiking deconvolution (e). Traces (f), (g), and (h) result when single-step deconvolution is applied to the input trace (c), using the operator lengths n and prediction lags α , as indicated.

Predictive deconvolution for multiple suppression (Figs. 2-64 and 2-65)

Use of two-step deconvolution process with different n and α

- Step 1: Predictive decon with large gap removes multiple
- Step 2: Spiking decon with gap=2 ms.
- Can also do in reverse order
- With single step with very large n for single primary reflector (Fig. 2-64)
- Single step decon generally not adequate for multiple primary reflectors (Fig. 2-65fgh)



Field examples of deconvolution



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Use of autocorrelogram to design decon operators that improve imaging of reflectors

- set window for optimizing parameters for reflectors rather than noise or other types of arrivals (e.g. refractors, guided waves) (e.g. Fig. 2-66c). But problems if length of autocorrelogram is too short (e.g. Fig. 2-66d)
- set n to include length of W and reverberations (e.g. 80-160 ms; Fig. 2-67)
- set lag small for spiking decon. Higher values will give more reverberations (Fig. 2-68)

Reflectors at 1.1, 1.35, 1.85 and 2.15 sec
 n = length of wavelet; α = short lag for spiking decon

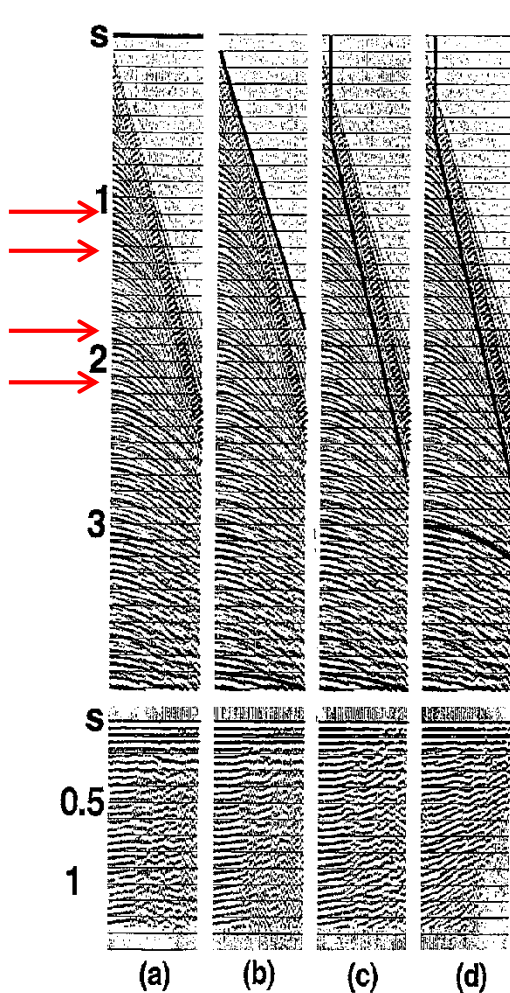


FIG. 2-66. An autocorrelation window test used to design deconvolution operators. The solid bars indicate the window boundaries. The entire 6-s length was included in a. The autocorrelation plots are displayed beneath the records.

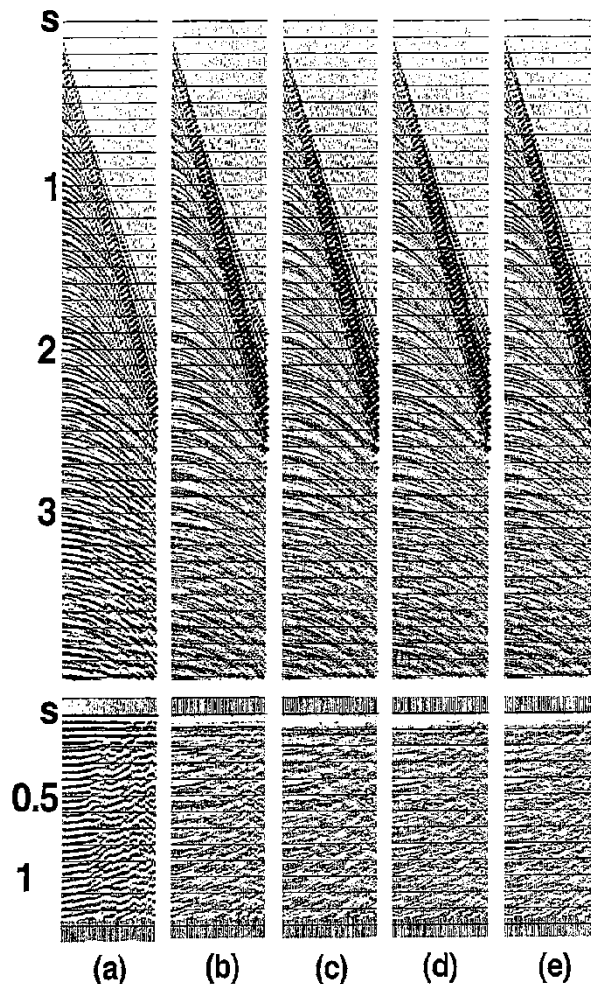


FIG. 2-67. Test of operator length. The corresponding autocorrelation is beneath each record. The window used in autocorrelation estimation is shown in Figure 2-66c. (a) Input gather. Deconvolution using prediction lag = 4 ms (spiking deconvolution), 0.1 percent prewhitening, and prediction filter operator lengths (b) 40 ms, (c) 80 ms, (d) 160 ms, (e) 240 ms.

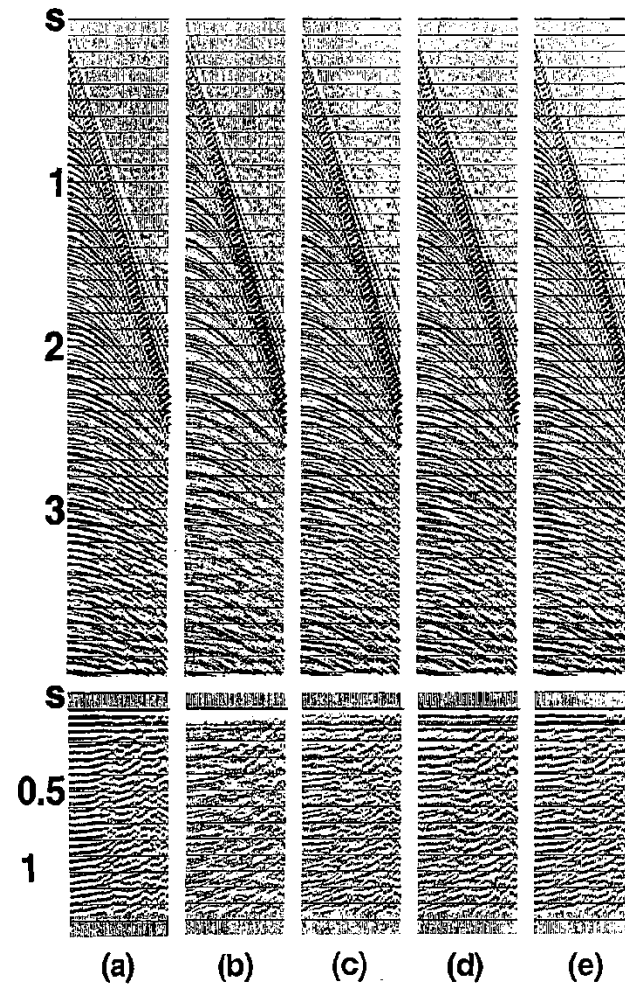
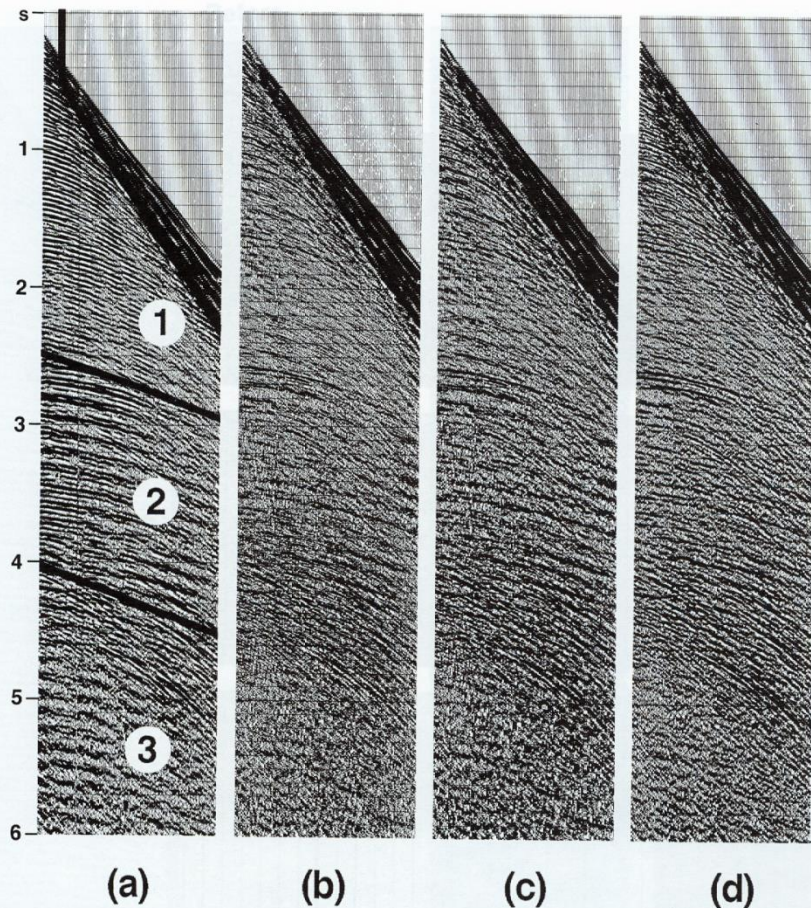
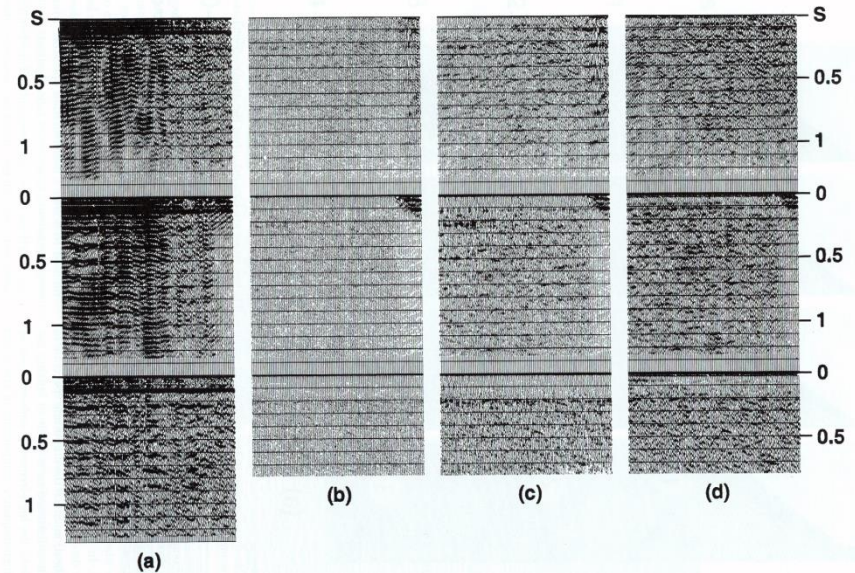


FIG. 2-68. Test of prediction lag. The corresponding autocorrelation is beneath each record. The window used in autocorrelation estimation is shown in Figure 2-66c. (a) Input gather. Deconvolution using prediction filter operator length = 160 ms, 0.1 percent prewhitening, and prediction lags (b) 12 ms, (c) 32 ms, (d) 64 ms, (e) 128 ms.

- Use of multiple windows used to account for non-stationarity of W as it travels deeper into sub-bottom (Fig. 2.6-4)
- Note differences in autocorrelogram between different sections (Fig. 2.6-5)



2.6-4. Three-window deconvolution. The solid bars indicate window boundaries. With data from each window, a deconv operator is designed and applied to the data in that window. The operators are blended across the window boundary. Input gather. Deconvolution using operator length of 160 ms and prediction lags of (b) 4 ms, (c) 12 ms, (d) 32 ms.



2.6-5. Autocorrelograms associated with the records shown in Figure 2.6-4. Note the difference in character between windows 1, 2, and 3 (from top to bottom in the far left panel). Windows are indicated in Figure 2.6-4a.

Signature processing (Figs. 2-75 and 2-76)

- Used when approximate signature of W is known. In this case W is split into two parts: a known wavelet (recorded in the water at far field) and the unknown part due to propagation within the sub-bottom and the recording system. Depends on accuracy for recording of W .
- A shaping filter can be used to produce minimum phase from the known part of W followed by spiking decon (Fig. 2-75)
- Alternate is to produce spike and then reduce ringing by predictive decon. Compare Fig. 2-75c,d,e to Fig. 2-76c,d,e. Since original W was not minimum phase these results should be better than previous result using decon of unknown W (Fig. 2-67d). What do you think?

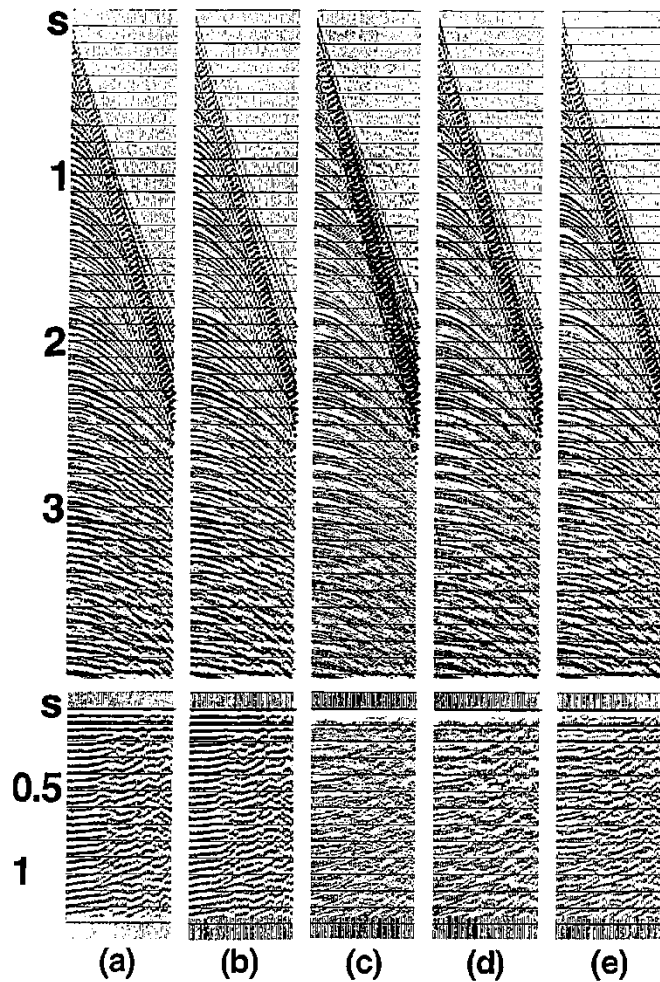


FIG. 2-75. Signature processing. A shaping filter is designed to convert the recording signature $[s(t)]$ in equation (2.31)] to its minimum-phase equivalent and apply it to the input record (a). Note that the output (b) has the same bandwidth as the input (a). The output (b) then has been processed by predictive deconvolution using operator length = 160 ms. and prediction lags (c) 4 ms (spiking deconvolution). (d) 12 ms. (e) 32 ms.

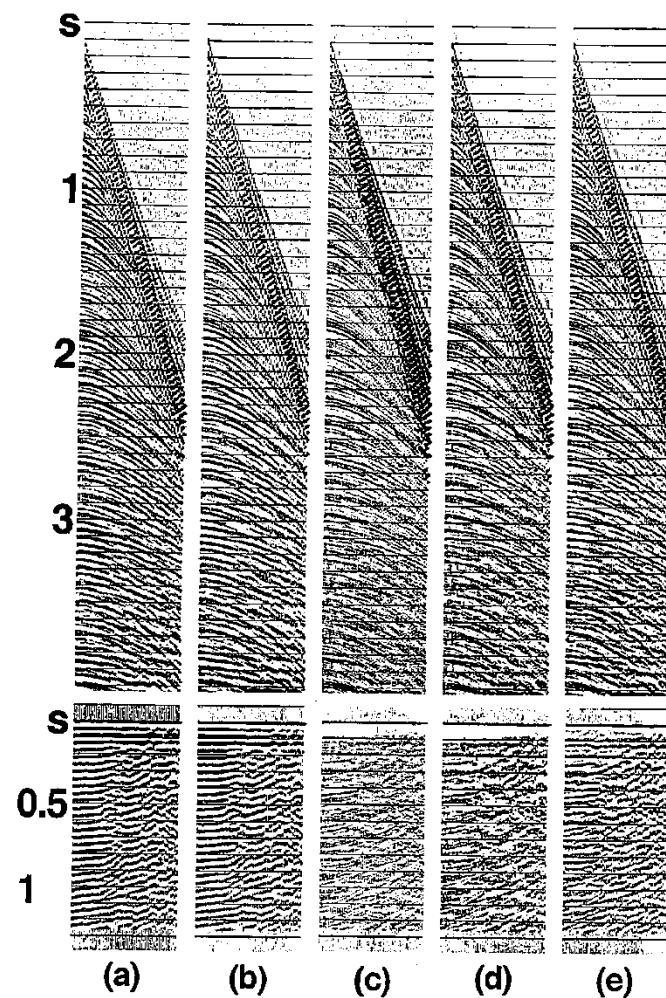


FIG. 2-76. Signature processing. A shaping filter is designed to convert the recorded signature $[s(t)]$ in equation (2.31)] to a spike and apply it to the input record (a). The output (b) then has been processed by predictive deconvolution using operator length = 160 ms and prediction lags (c) 4 ms (spiking deconvolution), (d) 12 ms. (e) 32 ms. Note that the output from signature processing (b) does not appear to have white spectrum; the $w(t)$ component [equation (2.31)] still needs to be removed.



Decon after stack (Fig. 2.6-14)

- Because assumptions of decon are never met in practice, the decon before stack (DBS) cannot produce an exact spike. Predictive decon applied to CMP stack may be more successful in removing multiples since noise is reduced by stacking.
- Generally followed by band-pass filtering to reduce noise that has been enhanced by decon

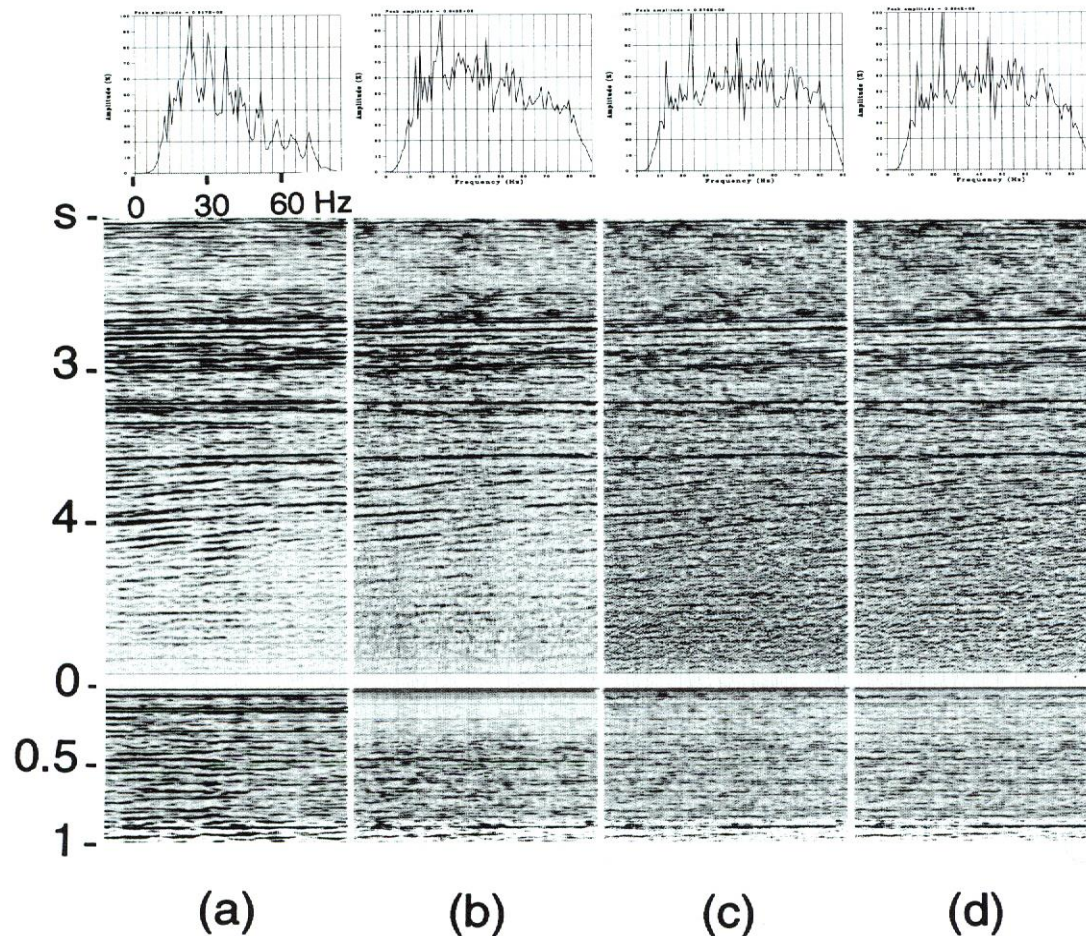


FIG. 2.6-14. (a) A portion of a CMP stack and its amplitude spectrum averaged over the CMP range (top) and autocorrelogram (bottom); (b) after spiking deconvolution, (c) followed by time-variant spectral whitening, and (d) time-variant filtering.

